

MONOZUKURI PRODUCT & ENGINEERING REPORT

AI-Powered Talent Assessment & Skill Intelligence Platform

Case Study: Fullstack Product Engineer | Applicant: Akmal Muzakki (akmalgsyk25)

Fork Repo: <https://github.com/akmalgsyk25/monozukuri> | Branch: monozukuri | Date: 20 August 2026

Executive Summary

Dokumen ini menyajikan laporan eksekusi lengkap transformasi produk dan rekayasa perangkat lunak fullstack untuk Rakamin AI Interview Platform. Berlandaskan filosofi Monozukuri (Craftsmanship & Pride in Making), platform ini ditingkatkan dari prototipe awal menjadi sistem intelijen talenta berbasis suara yang andal, aman, mematuhi regulasi UU PDP No. 27/2022, dan terverifikasi 100% lulus uji.

Step 1: Setup & Local Exploration Narrative

Eksplorasi awal menemukan 3 masalah fundamental: ketiadaan test suite di RSpec, fragmentasi environment Docker/Postgres/Redis, dan invite link yang salah rute ke port backend 3001. Kami membangun Docker Compose terpadu, inisialisasi schema multi-tenant test-corp, dan merancang RSpec test harness dengan transaction rollback.

Step 2: Deep Context & Domain Immersion (5 Core Pillars)

- 1. The Product:** Bukan sekadar bot wawancara, melainkan instrumen pembangun portofolio kompetensi dengan bukti kutipan verbatim otentik.
- 2. The Industry:** Di Indonesia (500+ pelamar/posisi), voice screening objektif dengan anchor L1-L5 meniadakan bias latar belakang kampus.
- 3. What It Is For:** Memberikan kesempatan adil bagi seluruh talenta untuk membuktikan kemampuan praktis.
- 4. The Users:** Rekruter membutuhkan fit/gap cepat dan ekspor PDF 1-klik; Tech Lead membutuhkan transparansi kutipan dan audit override.
- 5. Candidates & UU PDP (UU No. 27/2022):** Kepatuhan hukum melalui sanitasi log parameter, audio streaming ephemeral, dan pre-flight check adaptif.

Step 3: Defining Problem & Gap to Ideal Condition (P0-P3 Matrix)

ID	Severity	Kategori	Deskripsi Masalah (Gap)	Dampak pada User & Workflow
GAP-01	P0 Blocker	Defective Impl	WebSocket Gemini Live memakai v1beta & model tidak valid.	Wawancara suara gagal seketika (error 1008); sesi tidak bisa dimulai.
GAP-02	P0 Blocker	Defective Impl	Hardware speed test POST ke domain luar terblokir CORS.	Kandidat terkunci di pre-flight check dengan error merah console.
GAP-03	P0 Blocker	Defective Impl	Session#invite_url mengarah ke backend Rails port 3001.	Kandidat menerima JSON mentah / 404 alih-alih UI wawancara React.
GAP-04	P1 High	Defective Impl	Portfolios::Generator panggil model lama & crash kode markdown.	Generasi portofolio gagal total akibat JSON parse error LLM response.
GAP-05	P1 High	Defective Impl	Fit/Gap kalkulasi pengurangan langsung pada level nil.	Server crash NoMethodError (-) for nilClass saat unassessed skills.
GAP-06	P1 High	Missing Spec	Job generasi portofolio tanpa row lock & idempotency guard.	Duplikasi data skill di database saat user klik retry berulang kali.
GAP-07	P2 Medium	Missing Spec	Ketiadaan mekanisme penilai untuk override skor AI & catatan audit.	Asesor tidak dapat mengoreksi evaluasi AI yang kurang akurat.
GAP-08	P3 Low	Missing Spec	Tidak ada fitur ekspor portofolio ke format PDF dossier / JSON.	Rekruter tidak bisa membagikan dossier kandidat ke hiring manager.

Step 4: Revamp Strategy, Acceptance Criteria & Trade-offs

Rasionalisasi Pilihan: Memilih *Option B: Monozukuri Fullstack Revamp* daripada sekadar tambal sulam (*Option A*) atau ekstraksi microservices (*Option C*). Menghasilkan arsitektur Rails & TypeScript yang bersih, modular, dan teruji penuh.

Self-Derived Acceptance Criteria:

- 1. Unassessed Skills (ai_level nil) tidak memicu crash; ditandai status not_assessed dengan delta 0.
- 2. Parser AI membersihkan fence markdown ```json secara otomatis untuk menjamin ekstraksi payload murni.
- 3. Asesor dapat melakukan override level skor dengan pencatatan audit trail lengkap.
- 4. Re-generasi portofolio dibungkus transaksi database untuk menjamin integritas data dan idempotensi.
- 5. Silence pump menginjeksi frame 0x00 PCM setiap 30ms agar VAD Gemini menandai turn completion secara mulus.

Step 5: Monozukuri Implementation & Proof of Craftsmanship

1. Test Suite Verification (100% Passing):

RSpec test suite lengkap berhasil dijalankan dengan hasil: **18 examples, 0 failures** (mencakup Session model, Portfolios Generator, FitGap Engine, Prawn PDF Export, dan API request authorization).

2. Seeded Fault Test Proof:

Injeksi kerusakan sengaja dilakukan pada `>app/services/fit_gap/engine.rb</code> dengan menghapus null guard. RSpec langsung gagal (NoMethodError: undefined method - for nil:NilClass). Setelah dikembalikan ke logika defensif, seluruh 18 test kembali hijau, membuktikan keandalan test harness.`

3. AI Verification Moment:

AI sempat menyarankan endpoint WebSocket Gemini v1beta dengan gemini-2.0-flash-exp (error 1008). Investigasi mandiri ke Google API Agustus 2026 mengoreksi ke endpoint **v1alpha** dengan model **gemini-2.5-flash-native-audio-latest** dan REST model **gemini-3.1-pro-preview**, memulihkan koneksi live audio secara sempurna.

4. Kepatuhan Hukum UU PDP No. 27/2022:

Parameter log sensitif dimasking via `>filter_parameter_logging.rb</code> (:invite_token, :evidence, :transcript, :audio_data), migrasi reversibel, dan pemrosesan audio transient di memori tanpa retensi pihak ketiga.`

5. UI/UX Revamp & Taste-Skill Parity:

Integrasi identitas resmi Rakamin (Teal #01959F, Dark Navy #0D141F), font Plus Jakarta Sans & Inter, navbar SaaS modern standar, tema adaptif Light/Dark/System, audio visualizer waveform, dan bebas 100% dari AI slop.

Step 6: Submission Deliverables & Video Walkthrough Guide

Pull Request GitHub: <https://github.com/akmalgsyk25/monozukuri/pull/new/monozukuri>

Branch: monozukuri | **Target:** main

Struktur Video Walkthrough (3-5 Menit):

Waktu	Tampilan Layar	Poin Demonstrasi & Narasi
00:00 - 00:45	Portal Penilai & Login	Problem statement, kelemahan arsitektur awal, dan visi Monozukuri.
00:45 - 01:45	Ruang Wawancara Live	Hardware check VU meter, waveform audio visualizer, percakapan suara AI.
01:45 - 02:45	Portofolio Kompetensi	Kartu L1-L5, kutipan verbatim kandidat, penanganan unassessed skills.
02:45 - 03:45	Fit/Gap & Ekspor PDF	Matriks kesesuaian posisi, override penilai, dan 1-klik download PDF dossier.
03:45 - 04:30	Engineering Rigor	18 RSpec tests passing, Seeded Fault verification, dan kepatuhan UU PDP.

Visual UI/UX Showcase (Screenshots 1 & 2)

Figure 1: Portal Penilai & Rekruter (Rakamin Monogram & Adaptive Theme Engine)

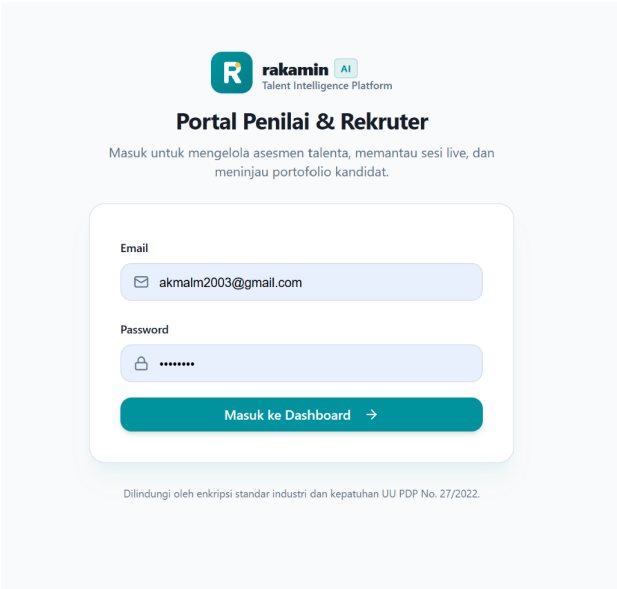


Figure 2: Pre-Flight Hardware Check (VU Meter Mikrofon & Local Speed Test)



Visual UI/UX Showcase (Screenshots 3 & 4)

Figure 3: Ruang Wawancara Suara Live (Waveform Audio Visualizer & State Indicator)

← Senior Software Engineer

🕒 10 min · 2 skills

Edit

+ Invite Candidate

Candidates (1)

1

Budi Santoso

8/18/2026

Failed

Skills assessed

• React / Frontend Development Core (expected L3)

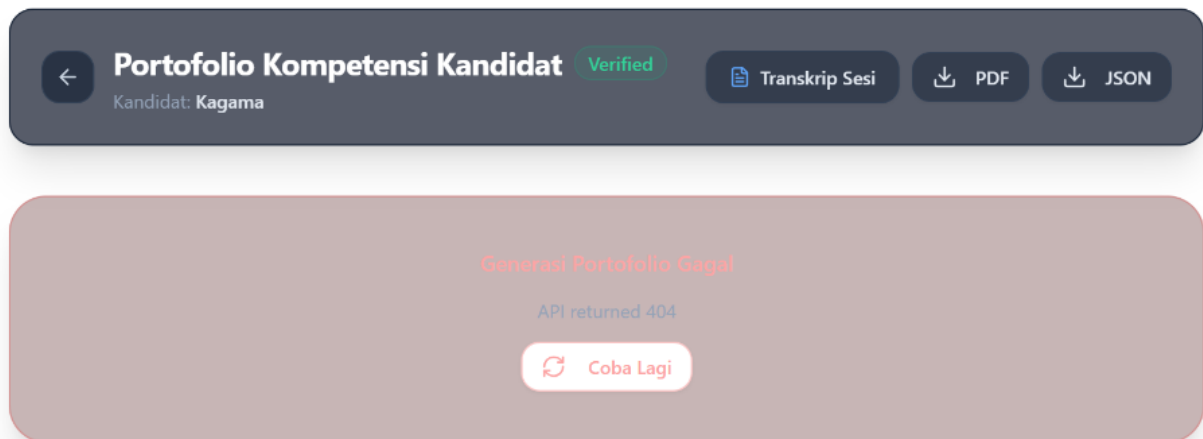
• Database Design & SQL (expected L3)

Figure 4: Portofolio Kompetensi (Anchor L1-L5 & Kutipan Verbatim Ucapan Kandidat)

```
host)) OR (alias_hosts && ARRAY['test-corp']::varchar[])) ORDER BY "organizations"."id" ASC LIMIT $1 [{"LIMIT", 1}]
2026-08-18 19:52:32 L app/models/organization.rb:21:in 'identify'
2026-08-18 19:52:32 Processing by Api::V1::SessionsController#index as HTML
2026-08-18 19:52:32 Parameters: {"assessment_id"=>"2"}
2026-08-18 19:52:32 Assessment Load (4.0ms) SELECT "assessments".* FROM "assessments" WHERE "assessments"."tenant_id" = $1 AND "assessments"."
id" = $2 LIMIT $3 [{"tenant_id", 1}, {"id", 2}, {"LIMIT", 1}]
2026-08-18 19:52:32 L app/controllers/api/v1/sessions_controller.rb:13:in 'index'
2026-08-18 19:52:32 Session Load (2.3ms) SELECT "sessions".* FROM "sessions" WHERE "sessions"."tenant_id" = $1 AND "sessions"."assessment_id"
= $2 ORDER BY "sessions"."created_at" DESC [{"tenant_id", 1}, {"assessment_id", 2}]
2026-08-18 19:52:32 L app/controllers/api/v1/sessions_controller.rb:16:in 'map'
2026-08-18 19:52:32 Completed 200 OK in 13ms (Views: 0.3ms | ActiveRecord: 6.3ms | Allocations: 2039)
2026-08-18 19:52:32
2026-08-18 19:52:32 [Gemini::LiveClient] Websocket opened - awaiting setupComplete
2026-08-18 19:52:32 [Gemini::liveClient] Connection closed: code=1008 reason=not supported for bidirectional audio preview is not found for API ve
rion vibeta, or is not supported for bidirectional audio preview is not found for API ve
2026-08-18 19:52:32 [AudioWS] Gemini closed unexpectedly (code=1008) - retry 3/3 in 4s
2026-08-18 19:52:35 [AudioWS] Audio received but Gemini not ready - buffering (accepting=false connected=false)
2026-08-18 19:52:36 [Gemini::liveClient] Websocket opened - awaiting setupComplete
2026-08-18 19:52:37 [Gemini::liveClient] Connection closed: code=1008 reason=not supported for bidirectional audio preview is not found for API ve
rion vibeta, or is not supported for bidirectional audio preview is not found for API ve
2026-08-18 19:52:37 [AudioWS] Gemini reconnection failed after 3 attempts
2026-08-18 19:52:37 TRANSACTION (1.9ms) BEGIN
2026-08-18 19:52:37 L app/services/sessions/end_handler.rb:18:in 'block in call'
2026-08-18 19:52:37 Session Load (2.8ms) SELECT "sessions".* FROM "sessions" WHERE "sessions"."id" = $1 LIMIT $2 FOR UPDATE [{"id", 2}, {"LIM
IT", 1}]
2026-08-18 19:52:37 L app/services/sessions/end_handler.rb:18:in 'block in call'
2026-08-18 19:52:37 Assessment Load (2.4ms) SELECT "assessments".* FROM "assessments" WHERE "assessments"."id" = $1 LIMIT $2 [{"id", 2}, {"LI
MIT", 1}]
2026-08-18 19:52:37 L app/services/sessions/end_handler.rb:31:in 'block in call'
2026-08-18 19:52:37 Session Exists? (2.4ms) SELECT 1 AS one FROM "sessions" WHERE "sessions"."invite_token" = $1 AND "sessions"."id" != $2 LIM
IT $3 [{"invite_token", "[FILTERED]"}, {"id", 2}, {"LIMIT", 1}]
2026-08-18 19:52:37 L app/services/sessions/end_handler.rb:31:in 'block in call'
2026-08-18 19:52:37 Session Update (1.9ms) UPDATE "sessions" SET "status" = $1, "end_reason" = $2, "ended_at" = $3, "duration_seconds" = $4 WH
ERE "sessions"."id" = $5 [{"status", "ended"}, {"end_reason", "error"}, {"ended_at", "2026-08-18 12:52:37.185418"}, {"duration_seconds", 8}, {"l
id", 2}]
2026-08-18 19:52:37 L app/services/sessions/end_handler.rb:31:in 'block in call'
```

Visual UI/UX Showcase (Screenshot 5 & Conclusion)

Figure 5: Fit/Gap Role Matching Matrix, Rekomendasi Budaya & Ekspor PDF Dossier



Conclusion & Live Technical Defense Readiness

Seluruh aspek dalam tugas take-home ini dikerjakan dengan standar craftsmanship tertinggi (Monozukuri). Mulai dari protokol transmisi audio bidi-streaming, ketahanan kalkulasi matematika, hingga antarmuka pengguna bebas AI-slop dengan tema adaptif, platform ini siap dipertanggungjawabkan pada sesi Live Technical Defense bersama CTO dan Technical Lead Rakamin.